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const int emgPin1 = A0; // Analog pin for EMG sensor 1
const int emgPin2 = A1; // Analog pin for EMG sensor 2

const float minValue = 0.0; // Minimum value for the y-axis (e.g., 0V)
const float maxValue = 0.15; // Maximum value for the y-axis (e.g., 5V)

const int sampleSize = 100; // Number of samples to collect
float emgData1[sampleSize]; // Array to store filtered data for sensor 1
float emgData2[sampleSize]; // Array to store filtered data for sensor 2

float alpha = 0.2; // Low-pass filter constant (adjust for smoothness)

void setup() {
    Serial.begin(9600);
    while (!Serial) {} // Wait for serial connection
    Serial.println("Starting EMG Data Collection...");
}

void loop() {
    // Collect and filter data

    for (int i = 0; i < sampleSize; i++) {
        // Read raw EMG data from sensors
        float raw1 = analogRead(emgPin1) * (5.0 / 1023.0); // Convert to voltage
        float raw2 = analogRead(emgPin2) * (5.0 / 1023.0);

        // Apply low-pass filter
        if (i == 0) { // Initialize the first value
            emgData1[i] = raw1;
            emgData2[i] = raw2;
        } else {
            emgData1[i] = alpha * raw1 + (1 - alpha) * emgData1[i - 1];
            emgData2[i] = alpha * raw2 + (1 - alpha) * emgData2[i - 1];
        }

        //min and max values for plotting
        Serial.print("Min:"); Serial.print(minValue); Serial.print("\t");
        Serial.print("Max:"); Serial.print(maxValue); Serial.print("\t");

        Serial.print("EMG1:");
        Serial.print(emgData1[i],4); //Serial.print(",");
        Serial.print("\t");
        Serial.print("EMG2:");
    }
}

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Serial.println(emgData2[i],4);

    delay(10); // Simulate sampling rate (adjust based on your EMG signal
requirements)
  }
}
```